

JungHyun An

Undergraduate Researcher, Seoul National University
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Education

Seoul National University <i>B.S. in Mechanical Engineering; Double Major in Electrical & Computer Engineering</i> GPA: 4.06 / 4.30	Seoul, Republic of Korea 2021–2027 expected
Korea Science Academy of KAIST <i>High School Diploma</i>	Busan, Republic of Korea 2018–2021

Research Interests

Robot learning for manipulation; generative action models; system-aware policy deployment; cloud/edge robotics.

Research Experience

Undergraduate Researcher, SNU Advanced Intelligent Systems Lab <i>Advisor: Prof. Hyun Jong Yang, Department of Electrical and Computer Engineering</i>	Jan. 2026–Present <i>Seoul National University</i>
- Conducting research on communication-aware multi-robot control in Sionna-based wireless simulation environments. - Studying resource allocation, planning, and control for multi-robot systems under communication constraints. - Contributing to image processing and segmentation-based perception for warehouse robotic systems.	
Undergraduate Researcher, SNU Robotics Laboratory <i>Advisor: Prof. Frank C. Park, Department of Mechanical Engineering</i>	Mar. 2025–Jan. 2026 <i>Seoul National University</i>
- Developed simulation datasets for generative motion policies using MuJoCo-based robotic environments. - Conducted independent research on geometry-aware flow matching for constrained robotic trajectory data. - Studied generative modeling approaches for robot trajectory learning, control, and precision-sensitive manipulation.	

Selected Publications and Manuscripts

Contamination-Aware Task Allocation for Cell-Free Cloud-Robotic AMR Warehouses JungHyun An [†] , Sehyun Ryu [†] , and Hyun Jong Yang. Submitted to <i>IEEE Internet of Things Journal</i> .	Submitted
Geometry-Aware Flow Matching for Robot Trajectory Learning Developing data-efficient training methods for flow-matching-based generative robot policies, with emphasis on exploiting low-dimensional geometry in constrained trajectory data for precision-sensitive manipulation. [†] Equal contribution.	Manuscript in preparation

Awards and Honors

1st Place, Digital System Design Final Project , Seoul National University Built the fastest CNN accelerator design on FPGA among 32 project teams.	Fall 2025
3rd Place, 2023 Cryptanalysis Contest Contributed to a three-person team by developing an algorithmic approach for elliptic curve cryptography analysis. Funded by the Defense Security Agency, Republic of Korea.	Sep. 2023

Selected Coursework

Control and Robotics: Mechanical System Modeling and Control; Control Systems I; Introduction to Robotics; Robot Vision.
Computation and Systems: Digital Logic Design and Laboratory; Computer Organization; Introduction to Data Communication Networks; Digital Systems Design and Laboratory.

Military Service

KATUSA Sergeant, Republic of Korea Army <i>Protocol Office, 2nd Infantry Division / ROK-US Combined Division</i>	Jul. 2022–Jan. 2024 <i>Mandatory Military Service</i>
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Technical Skills

Programming: Python, C/C++, MATLAB, Verilog
Packages / Tools: PyTorch, NumPy, SciPy, MuJoCo, robosuite, Sionna, Git